

Jacob McLaughlin

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Education

University of California, San Diego | BS in Computer Science – GPA: 3.92/4.00 Expected Jun 2028

- **Awards and Honors:** Tau Beta Pi Engineering Honors Society, Chancellor's Honor List
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Software Engineering & Construction, Discrete Structures

Experience

Research Intern | Collaborative Intelligence Systems Lab – Riverside, CA Jun 2026 – Present

- Reverse-engineered a commercial-grade differential-drive robot's undocumented interfaces for use on UCR's campus, gaining full control of sensor and control pipelines built around MPC-based obstacle avoidance
- Deployed a full end-to-end indoor autonomous navigation stack, building a custom LiDAR map with FAST-LIO, and achieving 95% goal-reaching success rate over 40 trials
- Built an outdoor GPS campus navigation system from scratch in Python/ROS with ENU projection, Dijkstra planner, and a pure-pursuit follower routing over a 45-building graph, completing routes of up to 0.25 miles
- Integrated OmniVLA-edge under the obstacle avoidance system to navigate unknown environments, with an 85% success rate for goals up to 5 meters

Projects

AI Short-Form Video Clipper | clipsifter.com Jun 2026 – Aug 2026

- Built and deployed a full-stack SaaS (Next.js on Vercel, Python/Celery worker on Railway) that turns long videos into captioned 9:16 clips via an async pipeline over Redis, Postgres, and Cloudflare R2
- Implemented word-synced karaoke captions and LLM moment selection and Groq-hosted Whisper transcription
- Engineered speaker-tracked reframing with OpenCV YuNet face detection and crop panning
- Built a Redis/Celery job queue with idempotent processing and Stripe billing on a per-user usage ledger

University Web Applications | ubi.ucrhighlanders.org, archery.ucrhighlanders.org Jan 2026 – Aug 2026

- Developed a Next.js web application within a team of 8 for Cyber @ UCR (and previously UCR's Archery Club and Blood Initiative), leveraging SSR to reduce initial load times by up to 70% and improve SEO
- Implemented a responsive, mobile-first UI using Tailwind CSS, improving usability for 50+ student members

LiDAR Autonomous Delivery Robot Apr 2026 – Jun 2026

- Developed the autonomy stack for a differential-drive robot designed for food delivery with 4 team members
- Implemented A* path planning and GPS-tracking to compute optimal, collision-free trajectories in real time
- Designed and integrated 2D LiDAR-based mapping and localization with ROS-based environment perception

Terminal-Based Crossy Road | github.com/BacoJaco/terminalcrossy Sep 2025 – Dec 2025

- Designed a replica of Crossy Road with 3 team members, leveraging SOLID principles in C++
- Deployed CI/CD pipeline using Google Test to ensure 99% build stability during active development

Leadership Experience

Project Lead | ACM @ UCR: Forge – Riverside, CA Apr 2026 – Jun 2026

- Led 10 members across 5 sub-teams to deliver an AI-powered surveillance system, coordinating Python and Next.js development to launch a functional prototype 2 weeks ahead of schedule
- Mentored team members on YOLOv8, Python multithreading, and MQTT to accelerate technical onboarding
- Oversaw the development of a Next.js dashboard to display live camera feeds and YOLOv8 detections

Lead Game Developer | GameSpawn – Riverside, CA Sep 2025 – Jan 2026

- Led a team of 3 developers to build a multi-level Unity game, using Agile sprints for on-time delivery
- Developed 7 interactive levels using C#, implementing core gameplay mechanics and navigation systems

Skills

Languages: C++, Python, C#, Typescript, SQL, HTML

Frameworks & Libraries: React, Next.js, Node.js, Tailwind CSS, ROS, NVIDIA HoloScan, Unity

Tools & ML/CV: Git, Claude Code, PyTorch, YOLOv8, OpenCV, MQTT, Google Test, CI/CD